

PraisonAI vulnerable to sandbox escape via ``print.__self__`` builtins module leak in ``execute_code`` (subprocess mode)

Report Type: Weekly · Generated: May 31, 2026 16:17 UTC · Coverage: May 29 – May 31, 2026

1	1	0	0
Total Advisories	Critical	High	Medium

Executive Summary

Summary ``execute_code()`` in ``praisonaiagents/tools/python_tools.py`` (v1.6.37, subprocess sandbox mode) can be fully bypassed using ``print.__self__`` to retrieve the real Python ``builtins`` module, from which ``__import__`` can be extracted via ``vars()`` and runtime string construction. This achieves arbitrary OS command execution on the host, completely defeating the sandbox. This is a ****novel bypass**** that survives all patches for CVE-2026-39888 (frame traversal), CVE-2026-34938 (str subclass), and CVE-2026-40158 (``type.__getattr__`` trampoline). --- ## Severity ****CVSS:3.1/AV:N/AC:L/PR:L/UI:N/S:C/C:H/I:H/A:H — 9.9 Critical**** --- ## Root Cause Three independent gaps in the AST-based security validation: ### Gap 1: ``__self__`` missing from ``_blocked_attrs`` In CPython, all built-in functions (C-level functions) have a ``__self__`` attribute that returns the module they belong to. The built-in functions in ``safe_builtins`` (``print``, ``len``, ``range``, etc.) are the *real* CPython bui

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Top Threat Advisories

High-priority advisories ranked by CVSS score, exploitability, and actor attribution.

Advisory Title	Severity	CVSS	EPSS	AI Summary
PraisonAI vulnerable to sandbox escape via `print.__self__` builtins module leak in `execute_code` (subprocess mode).	Critical	9.8	—	## Summary `execute_code()` in `praisonaiagents/tools/python_tools.py` (v

MITRE ATT&CK;® Coverage

ATT&CK; technique density score: 0.0 · Techniques observed: 0 · Tactics covered: 0

Threat Actor Intelligence

Attribution analysis across advisories in this report period.

Threat Actor	Advisory Count	Associated CVEs / Indicators
CDB-UNATTR-CVE	1	—

Actionable Recommendations

Prioritized defensive actions based on this period's intelligence.

1. Apply patches immediately for PraisonAI vulnerable to sandbox escape via `print.\_\_self\_\_` builtins module leak in `execute\_code` (subprocess mode).
2. Enable enhanced monitoring for related IOCs.
3. Review SIEM rules for associated MITRE ATT&CK; techniques.

Appendix — Report Metadata & Legal

Field	Value
Report ID	intel--dd2d68d72ddeeb26fb80394e
Report Type	Weekly
Platform	CYBERDUDEBIVASH SENTINEL APEX v143.0.0 GOD-MODE
Generated At	2026-05-31T16:17:22.908778+00:00
Customer Tier	PRO
Customer Email	—
TLP	TLP:CLEAR
STIX Version	2.1
ATT&CK Version	v15

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